

EXPERIMENT 7

COLUMNAR TRANSPOSITION CIPHER

AIM

To implement the Columnar Transposition Cipher using Python.

ALGORITHM

1. Read the plaintext and keyword.
2. Arrange the plaintext row-wise in a matrix.
3. Number the columns according to the alphabetical order of the keyword.
4. Read the columns in sorted-key order.
5. Display the encrypted text.

PYTHON CODE

```
import math

text = input("Enter Plain Text: ").replace(" ", "").upper()
key = input("Enter Key: ").upper()

col = len(key)
row = math.ceil(len(text) / col)

fill = row * col - len(text)
text += "X" * fill

matrix = []

k = 0
for i in range(row):
    matrix.append(list(text[k:k + col]))
    k += col
```

```
order = sorted(list(enumerate(key)), key=lambda x: x[1])
```

```
cipher = ""
```

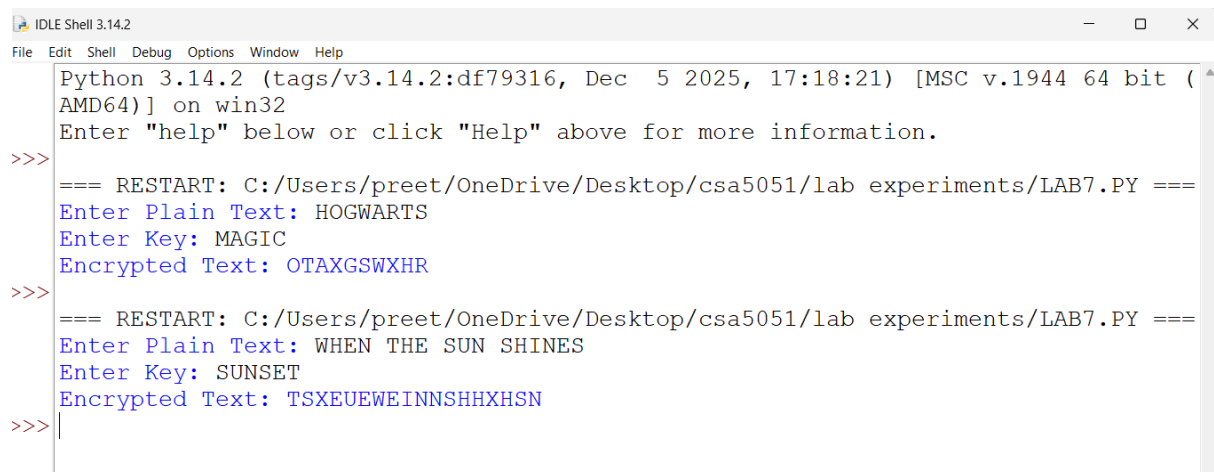
```
for index, _ in order:
```

```
    for r in matrix:
```

```
        cipher += r[index]
```

```
print("Encrypted Text:", cipher)
```

OUTPUT



```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB7.PY ===
Enter Plain Text: HOGWARTS
Enter Key: MAGIC
Encrypted Text: OTAXGSWXHR
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB7.PY ===
Enter Plain Text: WHEN THE SUN SHINES
Enter Key: SUNSET
Encrypted Text: TSXEUEWEINNSHHXHSN
>>>
```

RESULT

Thus the Columnar Transposition Cipher was implemented successfully using Python.